Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Moldover, Michael R. Ph.D.		
Project Title:	Acoustic Study of Critical Phenomena in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	GOVERNMENT	**Phone:**	301 975-2459
Organization Name:	National Institute of Standards and Technology		
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City:	Gaithersburg	**State:**	MD
Zip Code:	20899-8360	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	04/21/2004	**End Date:**	04/20/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Iek, Chanthy	**Contact Phone:**	216-433-3897
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Flight Program:			
Flight Assignment:	Note: grant renewed 4 yrs/new NAG#		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gillis, Keith (NIST) Zimmerli, Gregory (NASA Glenn Research Center)		
Grant/Contract No.:	NNC04IA20I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Using a double-Helmholtz acoustic resonator, also known as a Greenspan viscometer, we will measure the speed of sound and acoustic dissipation in a sample of xenon within 6 mK of the critical temperature. The speed of sound data will yield accurate measurement of the constant-volume heat capacity, and the quality factor of the resonator will be used to determine the bulk viscosity. The objective of this research is to measure the bulk viscosity of xenon near its liquid-vapor critical point using a novel acoustic resonator. The new measurements will be at frequencies that are a factor of 4000 lower than previous measurements, allowing us to extend the range of useful data a factor of 60 closer to the critical temperature. The bulk viscosity (or second viscosity) is the transport coefficient associated with dilational motion of a fluid. The bulk viscosity of a gas is often negligible in comparison to the shear viscosity. However, systems such as critical fluids or polyatomic gases may require many collisions before their internal degrees of freedom come into equilibrium with translational degrees of freedom, and the bulk viscosity can become much larger than the shear viscosity. Our measurements in microgravity will provide the most rigorous possible test of the theory for the bulk viscosity, i.e., the theory for the time-correlations of pairs of critical fluctuations. The quality-factor data will also test the theory of the thermal boundary layer in heretofore inaccessible ranges of frequency and fluid properties.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Progress was made in two areas: (1) Analysis of acoustic frequencies and halfwidth data for near-critical fluids, and (2) the use of convective stirring to simulate microgravity in the study of near-critical fluids. (1) The progress in the analysis and our data are described in a manuscript that has been accepted for publication: "Thermoacoustic Boundary Layers Near the Liquid-Vapor Critical Point" by K. A. Gillis, I. I. Shinder, and M. R. Moldover, Physical Review E,Vol 70, 021201 (2004). The abstract is: We measure and calculate the sound attenuation within thermoacoustic boundary layers between solid surfaces and xenon at its critical density rho_c as the reduced temperature t = (T – Tc)/Tc approaches zero. (Tc is the critical temperature.) Using the known thermophysical properties of xenon, we predict that the attenuation at the boundary first increases approximately as t^-0.6 and then saturates when the effusivity of the xenon exceeds that of the solid. [The effusivity is e = (rho·?C_P·lambda_T)^1/2, where C_P is the isobaric specific heat and lambda_T is the thermal conductivity.] The model correctly predicts (±1.0 %) the quality factors Q of resonances measured in a stainless steel resonator (effusivity = 6400 kg·K^-1·s^-5/2); it also predicts the observed increase of the Q, by up to a factor of 8, when the resonator is coated with a polymer (effusivity = 370 kg·K^-1·s^-5/2). The test data span the frequency range 0.1 kHz < f < 7.5 kHz and the reduced temperature range 10^-3 < t < 10^-1. We also predict that the thickness delta_T of the thermal boundary layer in the xenon decreases approximately as t^0.4 until 2·pi·f·gamma·zeta/(rho·c^2) » 0.5. (zeta is the bulk viscosity, gamma is the heat capacity ratio, and c is the speed of sound.) Still closer to Tc, delata_T becomes complex and its magnitude increases. These predictions concerning delta_T have not yet been tested. We deduce accurate values for the heat capacity C_V and thermal conductivity lambda_T for xenon in the range 10^-3 < t < 10^-1. (2) The progress in simulating microgravity is described in a draft manuscript that has been prepared for Physical Review E. The abstract for this draft manuscript is: We use a heater to force convective stirring within an acoustic resonator filled with near-critical xenon. The resonator had a volume of 20 cm^3, a complicated shape, and a maximum height of 47 mm. With convective stirring, the apparent equilibration time was reduced from several hours to less than 1 minute and the apparent temperature resolution was reduced from 100 mK to ~1 mK. This stirring technique enables high-resolution acoustic measurements of thermodynamic and transport properties at low frequencies close to the critical point.
Bibliography Type:	Description: (Last Updated: 09/16/2010)
Articles in Peer-reviewed Journals	Gillis KA, Shinder II, Moldover MR. "Thermoacoustic boundary layers near the liquid-vapor critical point." Physical Review E70, 021201, in press 2004. , Aug-2004